

BIG DATA ANALYTICS & MACHINE LEARNING IN THE REAL WORLD

May 2024

Big Data Analytics

DATA STORYTELLING

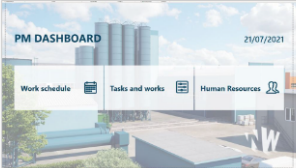
https://community.fabric.microsoft.com/t5/Data-Stories-Gallery/bd-p/DataStoriesGallery?sortBy=kudos

Microsoft | Fabric Community Forums Galleries Ideas User Groups Updates Blog Learning Resources

Data Stories Gallery

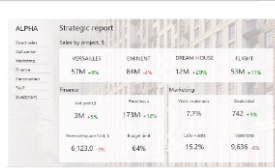
Recently Posted Featured **Top Kudos** All Data Stories

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Construction management

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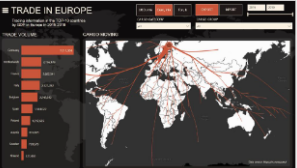
Property developer's sales analytics

alexkolokolov 537



Airport Authority Performance Summary

mmatey 334



Trade in Europe

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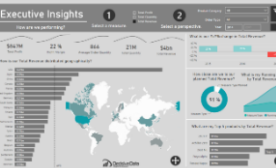
Global Super Store

mwadhwani 248



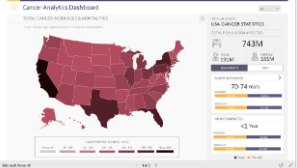
Climate Change is Real

ravalib 235



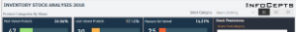
Executive Insights by Decisive Data

jolsby 226



Cancer Analytics

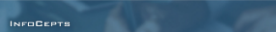
mmatey 213



Inventory Stock Analysis



Meal, to eat or not to eat?



Infocepts



Visual Storyboard

MEASUREMENT FRAMEWORK

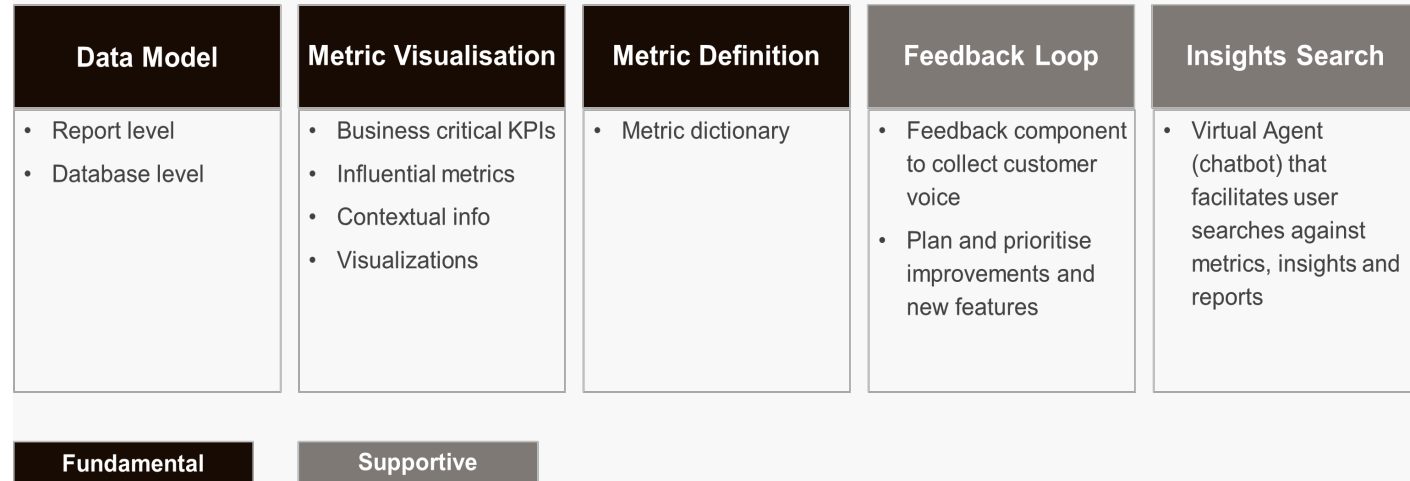
Tier 1: Business Outcomes	Audience: CMO or CDO Purpose: Assess Digital Commerce Results and Contribution to Business Goals.
Tier 2: Strategic Metrics	Audience: VP of Marketing and/or VP of Digital Commerce Purpose: Identify and Diagnose Issues Affecting Digital Commerce Results.
Tier 3: Operational Metrics	Audience: Category, Account, Platform or Program Director Purpose: Manage Marketing and Merchandising Based on Customer Behavior.
Tier 4: Tactical Metrics	Audience: Marketing Managers Purpose: Optimize Marketing Effectiveness for Impact on Digital Commerce Results.

AN EXAMPLE MEASUREMENT FRAMEWORK FOR E-COMMERCE

	Strategic Aim	Improving E-commerce growth and ROI			
	Key Result Area	Demand Generation	Digital Sales	Customer Experience	Brand & Reputation
	Strategic Metric	Digital Commerce Revenue Growth	E-commerce Revenue	Customer Lifetime Value	Loyalty CAST (Customer Satisfaction Score)
		Cross-sell / Upsell Ratio	E-commerce Revenue Profit Margin	Customer Acquisition Cost (CAC)	
		Sales Conversion Rate	E-commerce ReSales as a % of Total Sales		
Operational Metric	Operational Metric	Conversion Rate by Product Category	No. of Sales Transactions	Customer Acquisition Rate	Sentiment Rating
		Purchase Frequency	Avg Qty per Transaction	Retention / Renewal Rate	Rating by Product Category
		Repeat Visitor Rate	Avg Order Value (AOV)	CAC by Product Category	User Stickiness
		Visit to Page to Cart Conversion	Shopping Cart Abandonment Rate		
Tactical Metric	Tactical Metric	Click Through Rate (CTR)	Avg Time on Site	Registration Rate	Monthly Repeat Users
		Bounce Rate	Conversation by Channel	Engagement Rate by Product Category	Daily Avg Users
		Traffic by Channel	AOV by Marketing Campaign	Scroll Depth	
		Pageviews per Visit			
		Avg Dwell Time (login-to-checkout) by Product			
		Traffic by Segment			
		Traffic by Device			

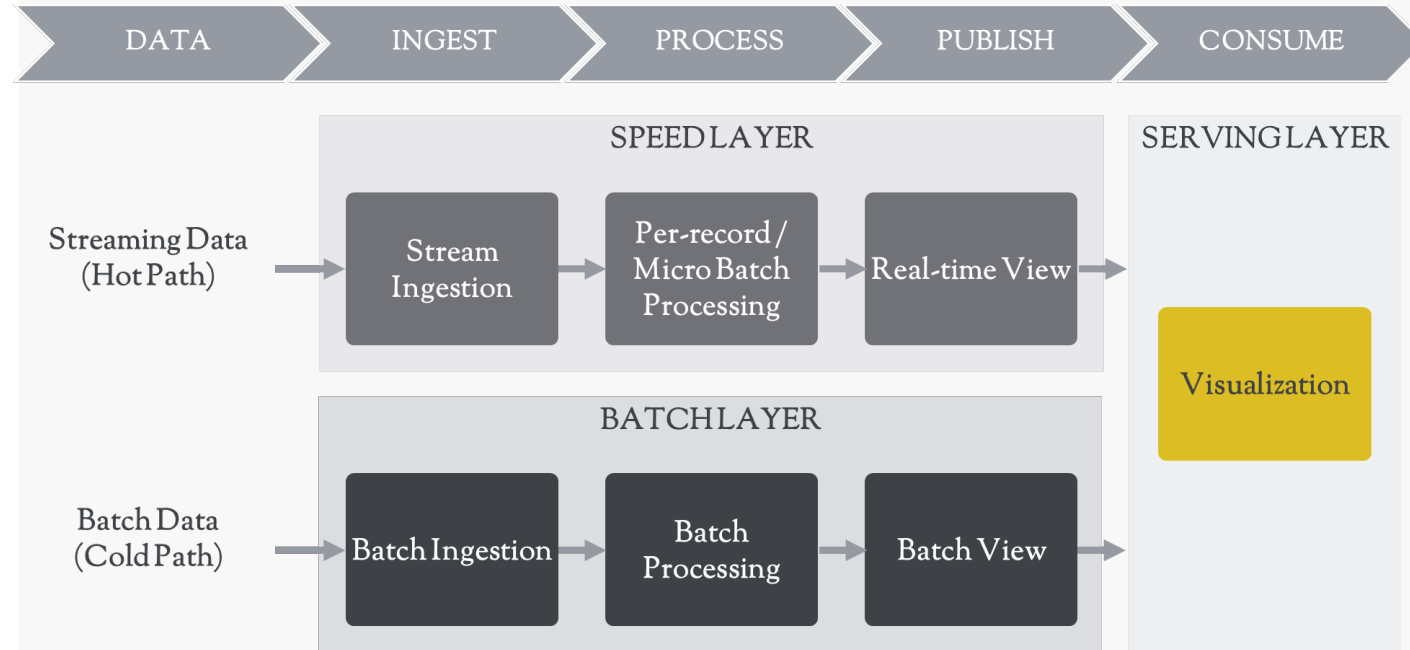
ANALYTICS SOLUTION FRAMEWORK

- Building blocks of a general analytics solution framework
- Visualisation - key to present metrics and KPIs
- Machine learning for creating data models



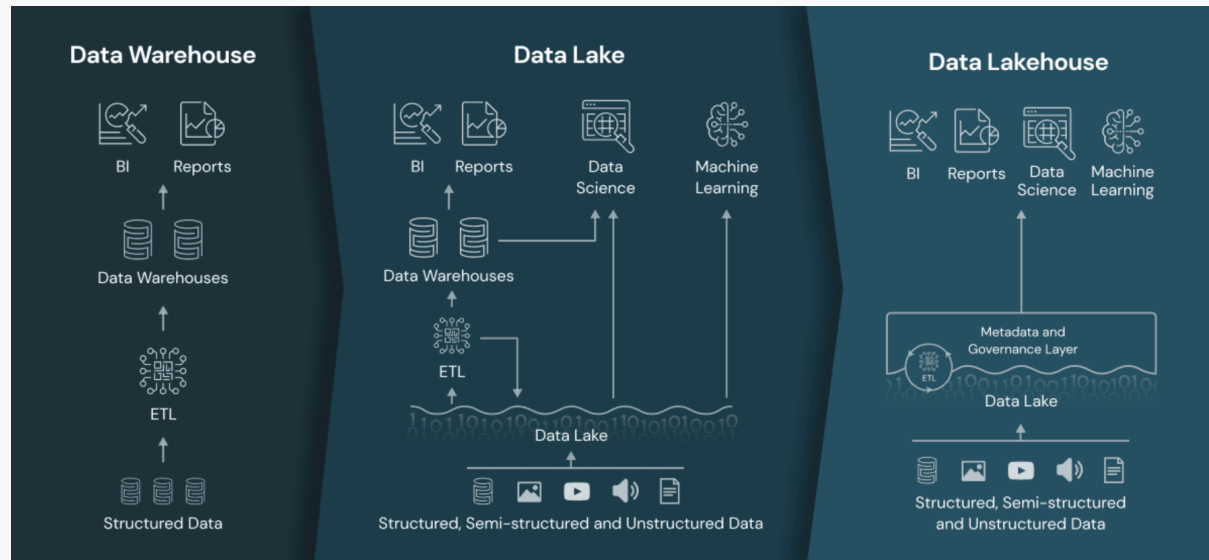
LAMBDA ARCHITECTURE

A classical approach
to big data
processing

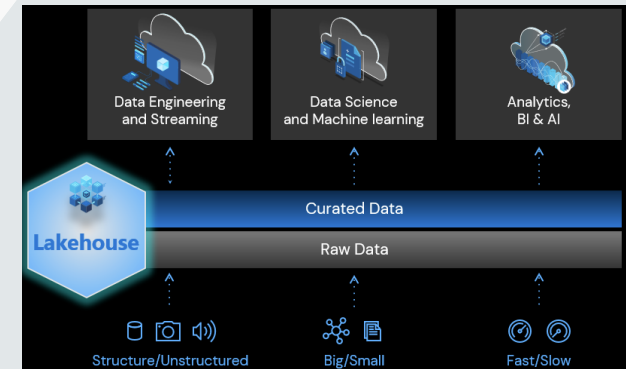


DATA LAKEHOUSE

- Lakehouse brings the best of data lake and data warehouse in a single unified data platform
- A single source of truth for data of all types and formats
- Integrated capabilities for storage, ETL/ELT, processing in parallel for scale, analytics, and machine learning



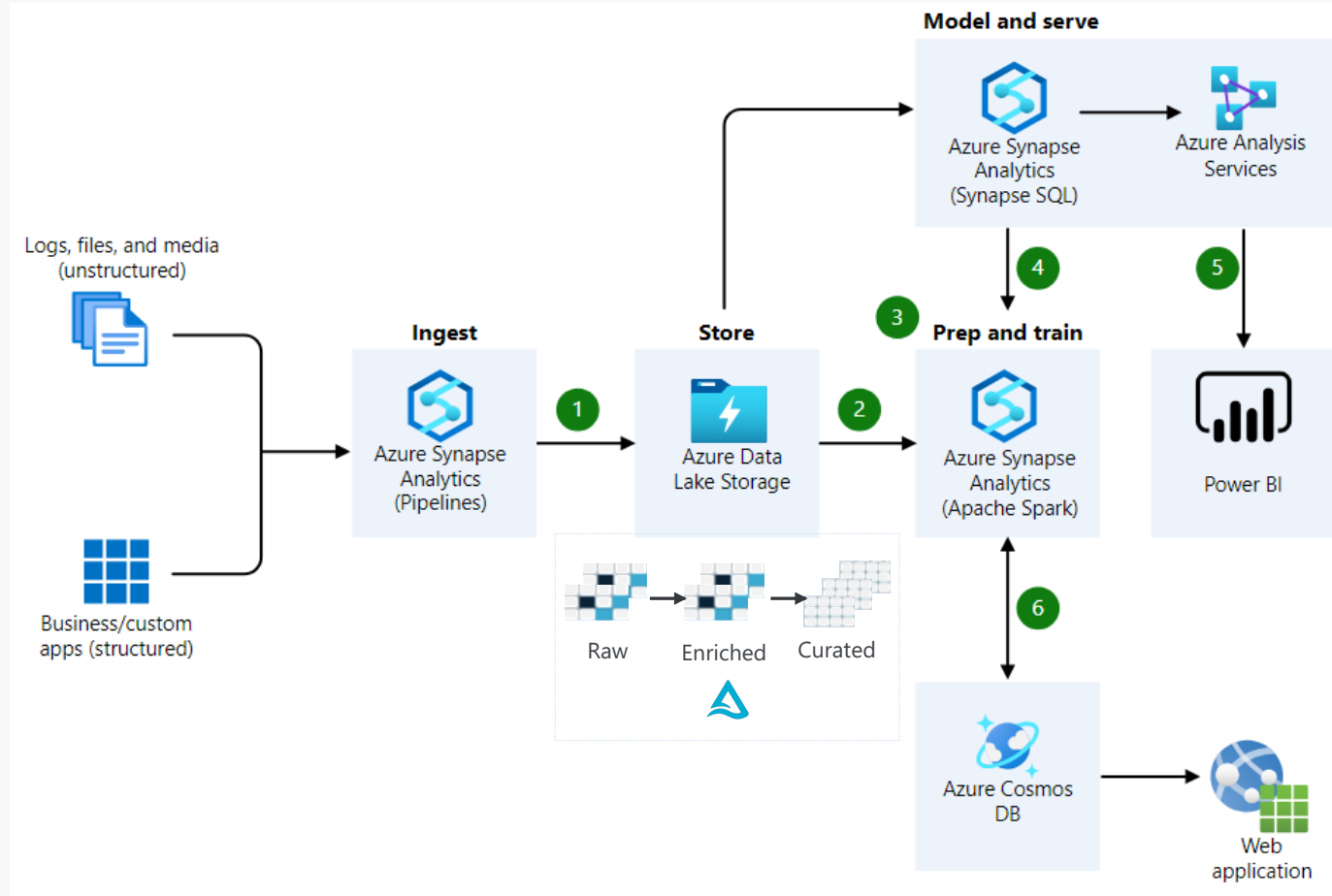
- There is no separate data warehouse and data lake. The data lake is the data warehouse.
- Delta Lake –is the secret sauce of the Lakehouse pattern and looks to resolve the limitations of data lake.



ADVANCED ANALYTICS ARCHITECTURE

Typical use cases:

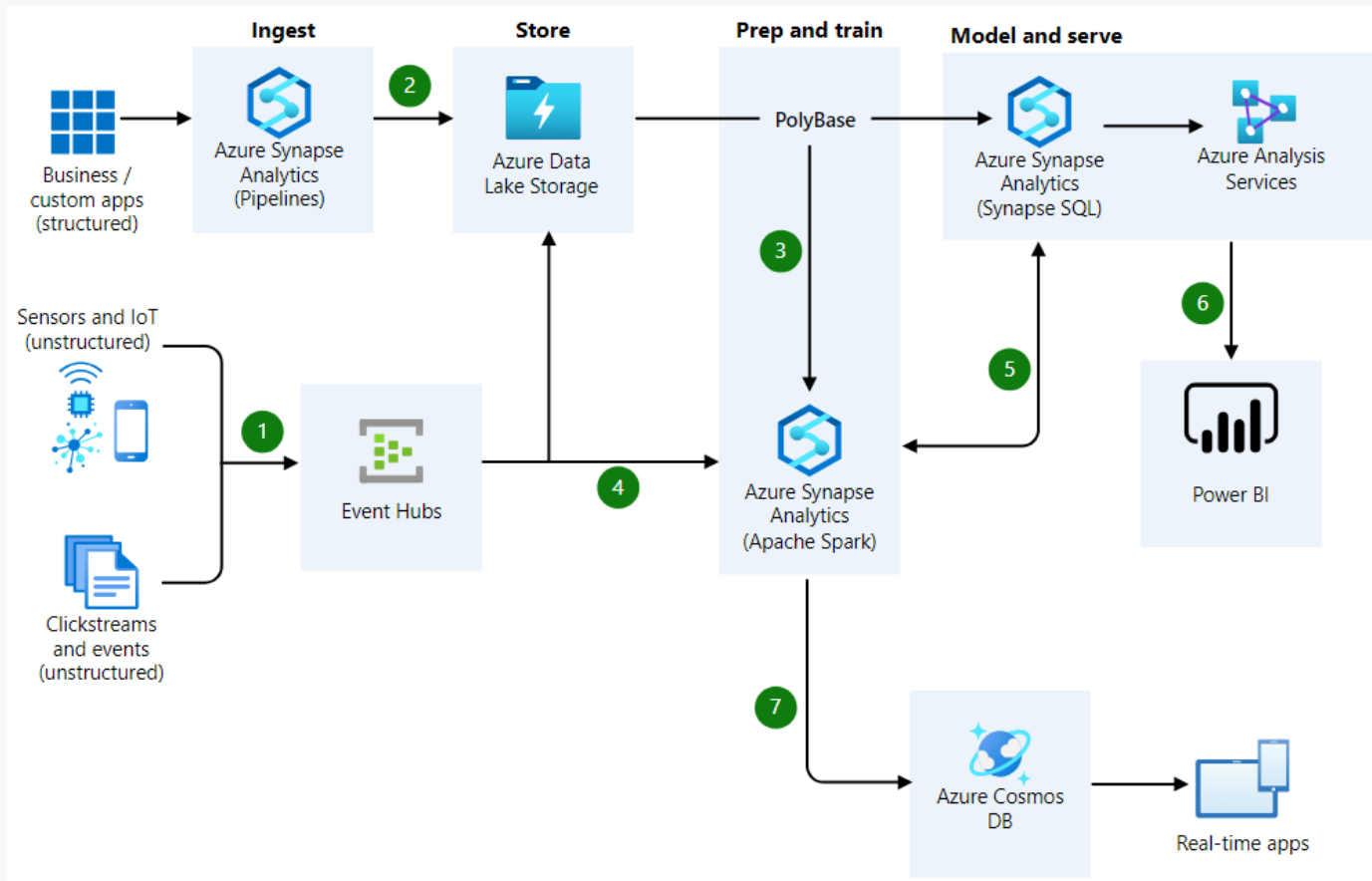
- Customer service
- Predictive maintenance
- Recommending products or services
- System optimization of everything from supply chains to data centre operations
- Product and services development



REAL-TIME ANALYTICS ARCHITECTURE

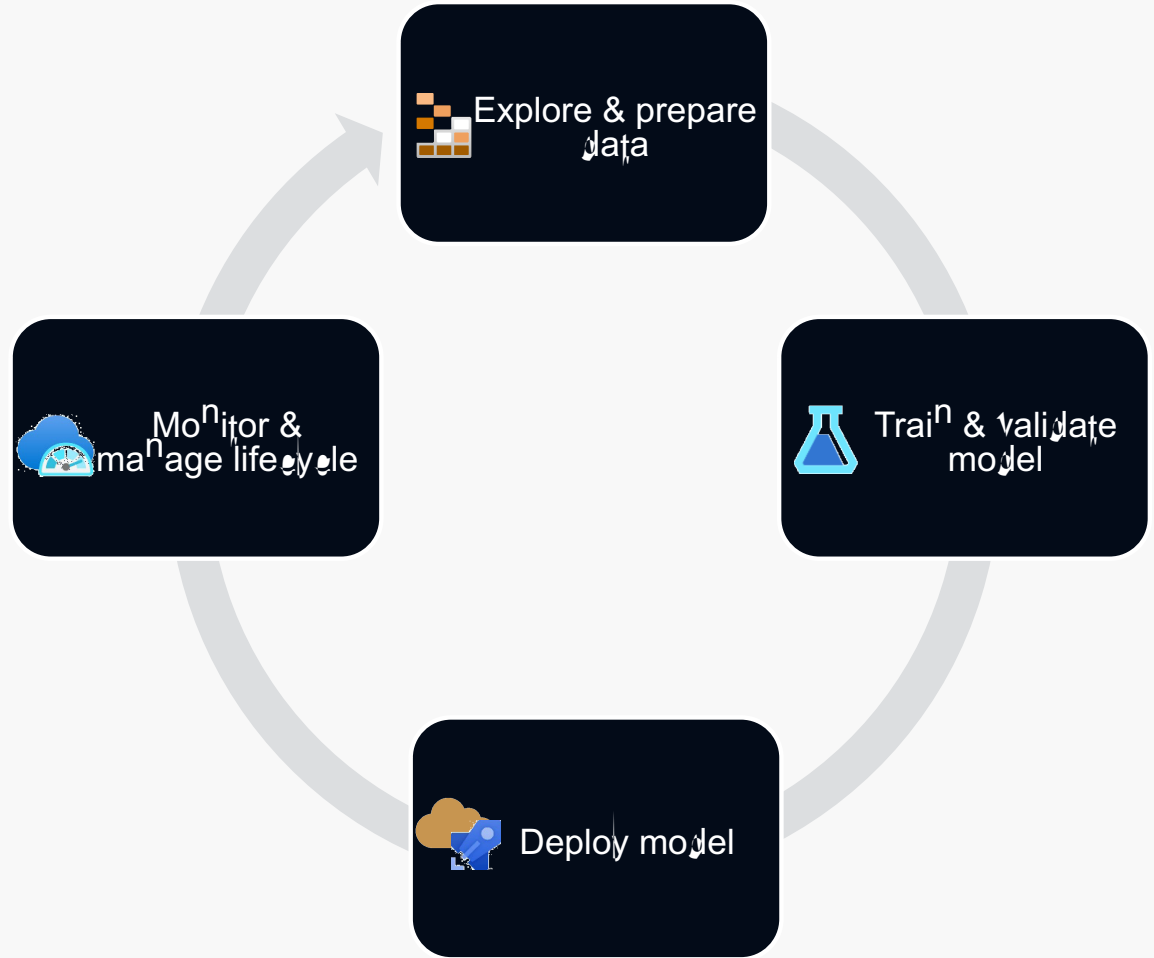
Typical use cases:

- Real-time traffic monitoring that uses sensor data to detect high traffic volumes
- a broker company that captures stock price over time to detect trends and optimise investment models
- Telemetry from sensors on industrial equipment to detect pending equipment failure and trigger alert notifications

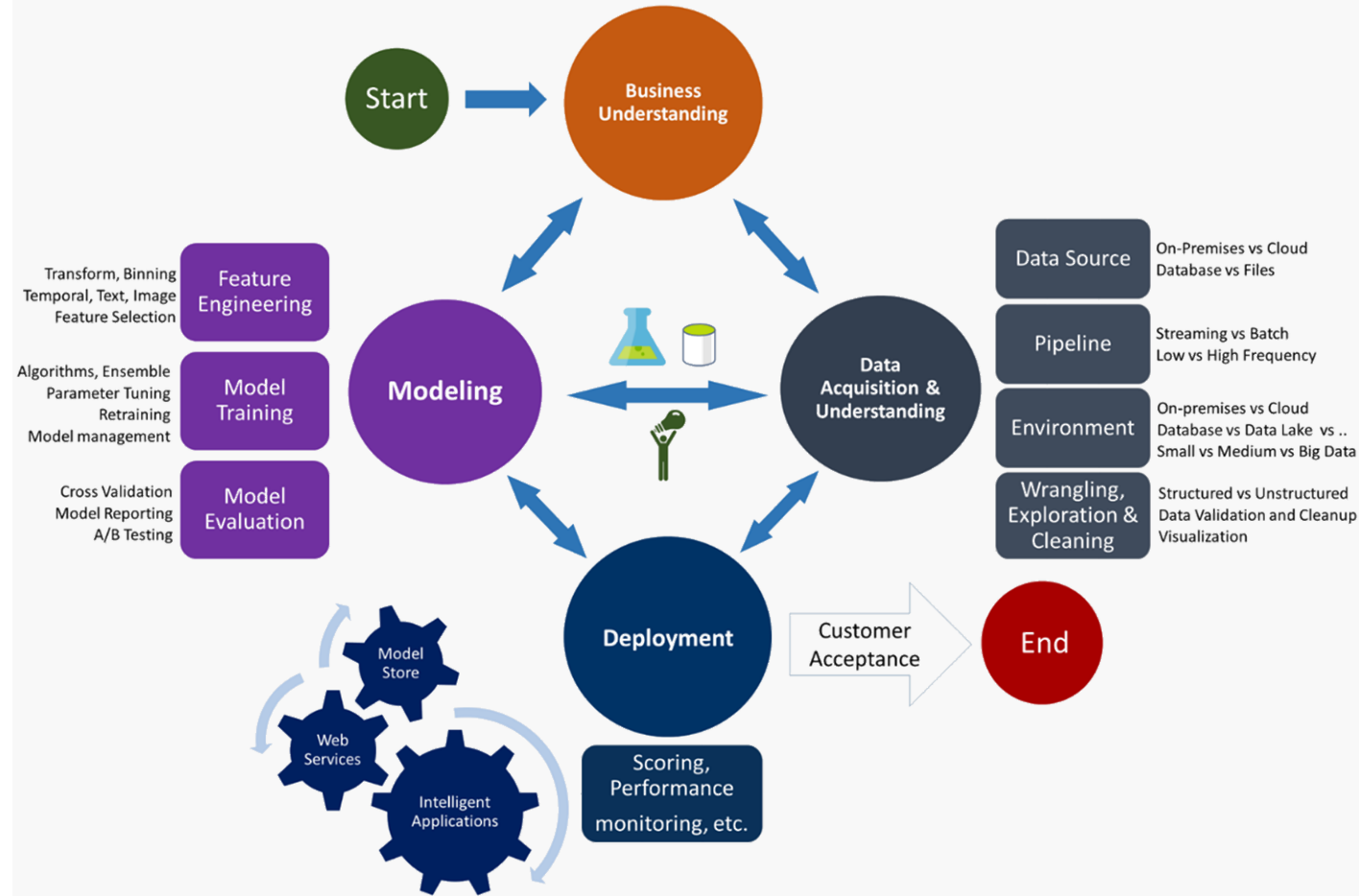


Machine Learning

MACHINE LEARNING LIFECYCLE: HIGH- LEVEL

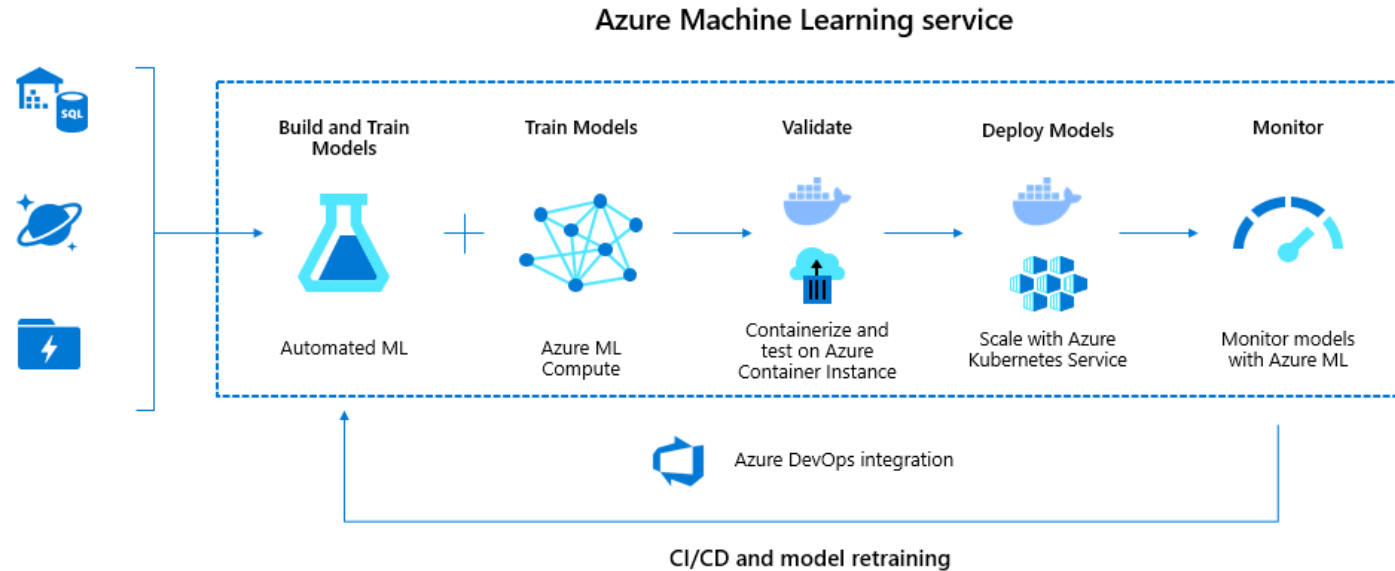


MACHINE LEARNING LIFECYCLE: DETAILED- LEVEL

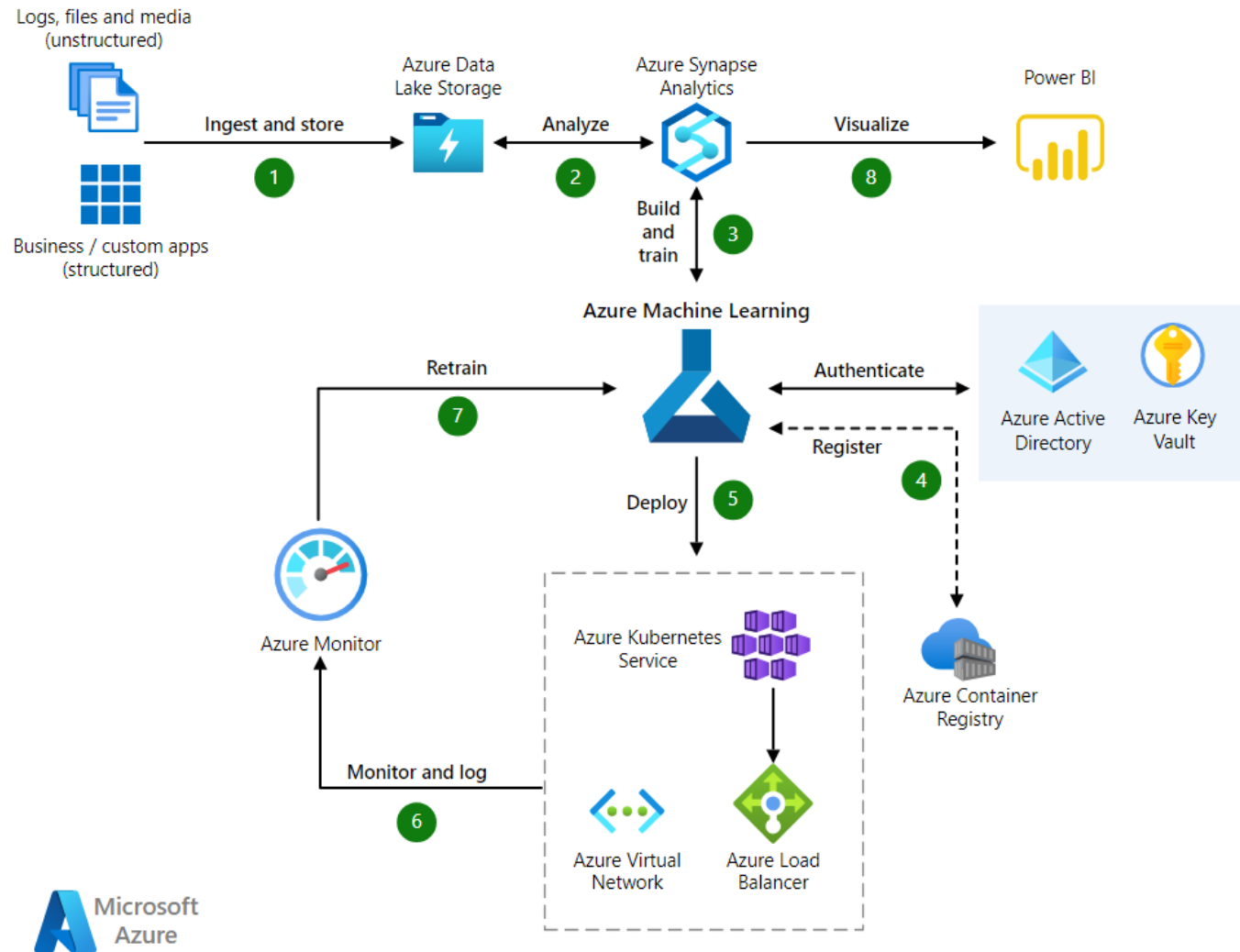


MACHINE LEARNING LIFECYCLE

Implementation with
Azure data services



AZURE ML ARCHITECTURE



MACHINE LEARNING IN HEALTHCARE



Source: [ML iⁿ Healthcare](#)

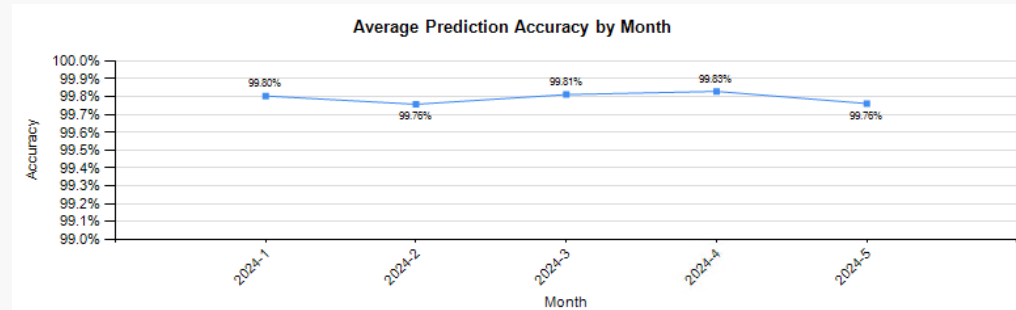
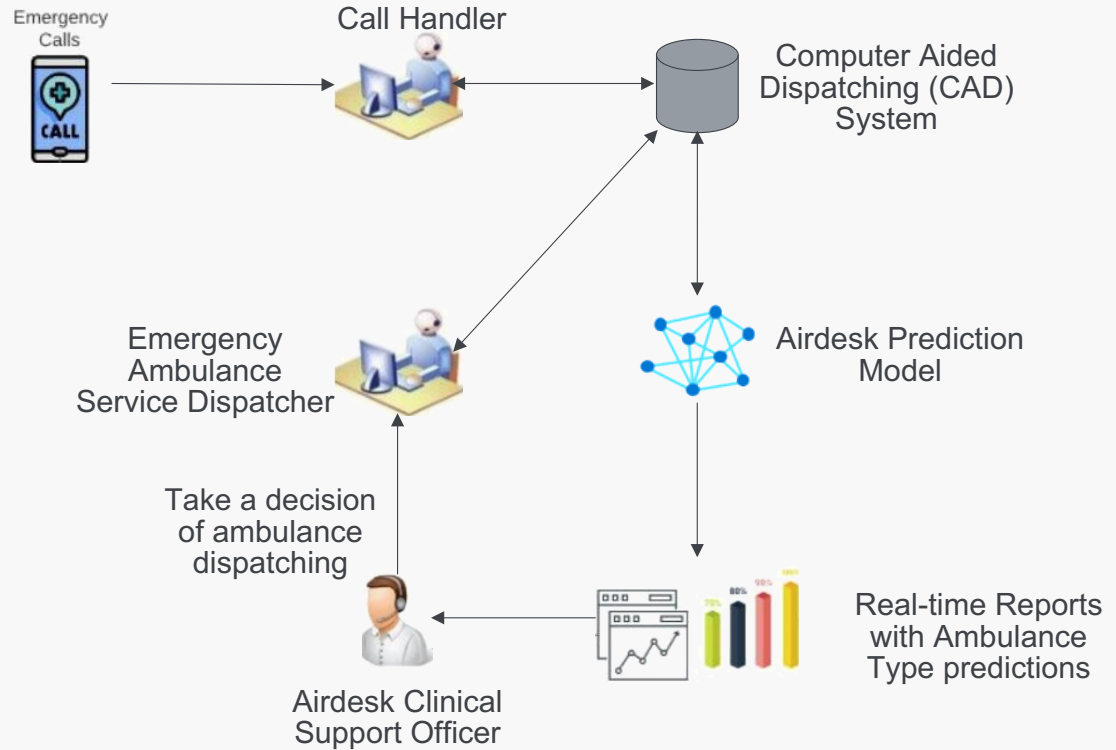
MACHINE LEARNING IN EMERGENCY HEALTHCARE

Use machine learning
to predict what type of
ambulance should be
sent to an incident



AIRDESK PREDICTION MODEL

- Ensemble learning
- Catboost decision tree
- 42 features
- Prediction accuracy: 99.2%
- Ongoing model maintenance plan



MACHINE LEARNING AT CALL CENTRE

Use machine learning to predict hourly call volumes, call wait time, and call handling time for a future week/month

